Energy Research & Social Science 134 (2026) 104631

Contents lists available at ScienceDirect

Energy Research & Social Science

journal homepage: www.elsevier.com/locate/erss

Original research article

An invisible threat? Exploring the physical and mental health risks of coal air pollution in the United States

Chien-fei Chen [a,e,*], Hang Shuai [b], R. Alexander Bentley [c,f], Lyudmyla Tsykalova [a], Zhenglai Shen [d]

[a] Department of Sociology, Anthropology and Criminal Justice, Clemson University, SC, USA
[b] Department of Electrical and Computer Engineering, Clemson University, SC, USA
[c] Department of Anthropology, University of Tennessee, Knoxville, TN, USA
[d] Buildings and Transportation Science Division, Oak Ridge National Laboratory, TN, USA
[e] Greater Resilience, Innovation and Transformation (GRIT) Labortaory, Charleston, SC, Clemson University, USA
[f] Center for the Dynamics of Social Complexity (DySoC), University of Tennessee, Knoxville, TN, USA

ARTICLE INFO

Keywords:
Fine particulate matter ($PM_{2.5}$)
Coal-fired power generation
Health risks
Energy burden
Energy justice
Environmental justice

ABSTRACT

Air pollution remains a significant environmental health risk in some regions of the United States, contributing to a wide range of health impacts, especially among socially and medically vulnerable populations. Historically, coal-fired power plants have been major sources of ambient fine particulate matter ($PM_{2.5}$); although their contributions have declined substantially in recent decades. Despite well-documented geographic declines in coal-related $PM_{2.5}$ exposure, less is known about persistent regional disparities and how they intersect with social determinants of health. This study examines associations between coal-related $PM_{2.5}$ exposure and health across U.S. counties in 2020, using nine indicators spanning mental and physical conditions. We find that a 1% increase in coal $PM_{2.5}$ exposure is associated with a 0.02% to 0.34% increase in the prevalence of several outcomes, including mental health disorders, chronic obstructive pulmonary disease (COPD), and diabetes. These associations vary markedly by region, with the strongest effects concentrated in the Southeastern United States. Additionally, high household energy burden amplifies the adverse mental and physical health impacts of coal $PM_{2.5}$ exposure. Persistent regional inequities underscore the need for targeted, place-based interventions to reduce ongoing local exposure and its health consequences.

1. Introduction

Air pollution is a major public health threat, contributing to an estimated 4 to 9 million premature deaths globally, with fine particulate matter ($PM_{2.5}$) responsible for over 80% of this burden [1,2]. $PM_{2.5}$ is a complex mixture of emissions from fossil fuel combustion, vehicles, power generation, agricultural and industrial processes together with natural sources such as desert dust [2–4]. Worldwide, coal emissions remain substantial [5]. In the United States (US), despite reductions in emissions over recent decades [6,7], coal-fired power plants remain a significant source of $PM_{2.5}$, producing sulfur dioxide (SO_2), nitrogen oxides (NO_x), and heavy metals such as lead and mercury [7,8]. These pollutants contribute to cardiovascular, respiratory, neurological diseases and increased overall mortality, along with other severe health effects in the U.S. [2,7,9] . Coal-generated $PM_{2.5}$ continues to pose regional and population-specific health risks [9] and represents a well-established threat to public health (e.g. [10]).

While socioeconomic determinants of pollution exposure have been documented [7], regional (geospatial) variations in the resulting physical and mental health effects, which may be persistent, remain unclear. Grounded in energy and environmental justice frameworks, this paper examines how coal-related fine particulate matter (coal $PM_{2.5}$), continues to affect physical and mental health in the U.S., despite substantial declines in overall emissions. The health benefits of reducing coal-fired power plant emissions of $PM_{2.5}$ (i.e., coal $PM_{2.5}$) have not yet been explored in terms of regional disparities or their interactions with the social determinants of health [9]. Using county-level data for U.S. counties in 2020, we investigate the associations between coal $PM_{2.5}$ exposure and nine health outcomes while also accounting for energy burdens (EBs) and key sociodemographic factors including race/

* Corresponding author.
E-mail address: chienfc@clemson.edu (C.-f. Chen).

https://doi.org/10.1016/j.erss.2026.104631
Received 17 September 2025; Received in revised form 20 February 2026; Accepted 24 February 2026
2214-6296/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

ethnicity, age, disability, and health insurance coverage. To assess the regional differences, we integrate spatial lag regression models with hot-spot analyses to identify geographic clustering of coal $PM_{2.5}$ and adverse health outcomes and to examine how these geospatial patterns vary across US regions.

Key findings indicate that even modest increases in coal $PM_{2.5}$ are associated with statistically significant increases in the prevalence of multiple chronic and mental health conditions, particularly in the Eastern U.S. High EBs, exhibiting the strongest overall associations with health outcomes, further amplify the health risks of coal-related air pollution. This study underscores the persistence and spatial unevenness of coal-related health and EBs, it also highlights the need for regionally targeted interventions to reduce pollution and support communities disproportionately affected by energy-related inequities [11,12]. The next section reviews the literature that provides the theoretical foundation for this study. Subsequent sections describe data sources, analytic strategy, results, and policy implications.

2. Literature review

2.1. Energy and environmental justice frameworks

This study integrates energy justice and environmental justice frameworks to examine how coal-related $PM_{2.5}$ exposure, household EBs, and social vulnerability jointly shape health risks. Energy justice provides a focused analytical lens for identifying and addressing inequities arising from energy production, distribution, and consumption, building directly on earlier environmental justice scholarship [5,13]. Early definitions emphasized *where* energy-related harms and benefits occur, recognition, *which* communities are marginalized, and *how* to mitigate these inequities [14,15]. Its core dimensions, including distributional, procedural, and recognitional justice, highlight that the risks and benefits of energy systems are unevenly shared, particularly among historically marginalized populations [11,16,17]. Energy justice is methodologically and geographically diverse, offering practical tools for assessing disparities in exposure, vulnerability, and access affordable and reliable energy services [18,19]. Within this framework, household EBs, defined as the proportion of household income spent on utility costs, functions as a key mechanism through which energy inequities translate into adverse health outcomes. High EBs can amplify health risks by constraining households' ability to maintain thermal comfort, access healthcare, invest in energy-efficient technologies, or adopt protective mitigation measures [18,20–22]. Because EB and energy-related environmental exposures are empirically measurable, energy justice is often viewed as more directly actionable than broader climate or environmental justice frameworks, serving as a decision-making tool to guide planners and policymakers toward safer, more affordable, and more sustainable energy systems [5,19,23].

Environmental justice, by contrast, is grounded in the pursuit of equitable environmental conditions, democratic participation, and recognition of historically marginalized communities, with deep roots in civil rights movements, human rights principles, and structural analyses of power [24]. Decades of environmental justice research demonstrates that air pollution exposure and the siting of hazardous facilities disproportionately burden low-income communities and communities of color. Building on this foundation, recent scholarship emphasizes that energy-related air-quality risks emerge across the entire energy life cycle, including resource extraction, electricity generation, waste handling, and consumption, extending environmental injustice beyond point-source pollution alone [25]. These cumulative processes often concentrate harms in rural, low-income, and politically marginalized communities [25].

The overarching goal of the energy justice framework is to assess whether all populations have equitable access to safe, affordable, and reliable energy within environments free from harmful pollution. When applied alongside environmental justice, this integrated approach highlights how structural disadvantages, including financial inequality, high EBs, social vulnerability, and historical patterns of disinvestment, interact to compound the physical and mental health impacts of coal-related $PM_{2.5}$ exposure across U.S. counties [26–28]. In sum, these frameworks provide a justice-oriented lens for explaining how energy systems, environmental exposures, and social inequities converge to produce uneven and persistent health risks. Although prior research shows that energy systems can produce and reinforce inequities, far fewer studies examine how environmental exposures intersect with economic energy burdens. In this study, we assess how coal-related $PM_{2.5}$ exposure and household energy burdens jointly shape population health outcomes across U.S. counties.

2.2. Public and population health impacts of coal-related $PM_{2.5}$

Over the past four decades, the Clean Air Act (CAA) and its subsequent amendments have played a central role in reducing air pollution from the U.S. power sector, particularly emissions of sulfur dioxide (SO_2) and nitrogen oxides (NO_x), which are key precursors to fine $PM_{2.5}$ [7]. Major regulatory milestones, including the 1970 Clean Air Act, the 1990 Clean Air Act Amendments, and market-based programs such as the Acid Rain Program and the Cross-State Air Pollution Rule, established emissions standards, cap-and-trade mechanisms, and interstate transport controls that fundamentally reshaped power plant operations and fuel choices [29]. As a result, the U.S. energy sector has reduced SO_2 emissions by approximately 67% while NO_x emissions have fallen by about 40% since the early 1980s [7]. These regulatory interventions have driven substantial nationwide reductions in ambient $PM_{2.5}$ concentration, with well-documented population health benefits, including reduced premature mortality and lower cardiopulmonary and respiratory morbidity. Yet it remains unclear how these CAA-driven improvements translate into health outcomes at the local level. Our analysis indicates that the largest remaining burdens are concentrated in areas with persistently elevated $PM_{2.5}$, especially the Southeast and Appalachia, indicating where additional reductions could yield the greatest health benefits.

Despite this progress, coal-fired power plants remain a major source of harmful $PM_{2.5}$ exposure. Coal-related $PM_{2.5}$ has been consistently linked to oxidative stress, systemic inflammation, and a wide range of adverse health outcomes [9,26]. Epidemiological estimates attribute nearly 460,000 premature deaths in the U.S. between 1999 and 2020 to $PM_{2.5}$ emissions from coal-fired power generation [9]. Coal-related $PM_{2.5}$ affects multiple organ systems through several biological pathways. Fine particles can penetrate deep into the lungs, enter systemic circulation, and reach the brain via the bloodstream or olfactory bulb, triggering neuroinflammation and oxidative damage [30]. Short-term exposure is associated with symptoms ranging from eye, nose, throat, and skin irritation to respiratory distress, chest tightness, wheezing, headaches, dizziness, and nausea. More severe acute outcomes include asthma exacerbations and cardiovascular stress [2,9,10,16,17,30,31]. Long-term exposure can result in cumulative damage to neurological, respiratory, reproductive, and cardiovascular systems [32], contributing to increased risks of respiratory and cardiovascular diseases, cancer, adverse birth outcomes, neurobehavioral disorders, depression, and premature mortality [30,31,33]. Additionally, $PM_{2.5}$ contributes to lung-function decline, chronic inflammation, oxidative stress, and disruption of proteolytic balance, and may act as a carrier for microorganisms that exacerbate COPD progression [4].

The cardiovascular impacts of $PM_{2.5}$ exposure are also well documented [32,34]. Inhaled particles induce oxidative stress and systemic inflammation, disrupt autonomic nervous system regulation, and may translocate directly into the bloodstream [32]. These mechanisms contribute to cardiac arrhythmia, coronary artery disease, cerebrovascular disease, heart failure, peripheral arterial disease, and venous thromboembolism. Further, $PM_{2.5}$ exposure alters cardiac autonomic activity, increases electrical instability, reduces myocardial perfusion,

and elevates the risk of ischemia, severe arrhythmia, and cardiovascular-related mortality. Long-term exposure has also been associated with accelerated cognitive decline and increased dementia risk in adulthood [35]. Systematic reviews additionally report increased suicide incidence following consecutive days of elevated $PM_{2.5}$, PM_{10}, or NO_2 exposure [36–39].

Children, pregnant women, older adults, and individuals with chronic diseases face increased risks [32]. Children are especially vulnerable to PM2.5 because they breathe air per unit of body weight while their lungs and immune defenses are still developing. Fine particles can penetrate deep into the airways, trigger oxidative stress and inflammation, slow in lung-function growth, and potentially lead to lifelong respiratory impairment [40]. Prenatal and early-life exposure to $PM_{2.5}$ exposures are associated with wide range of adverse outcomes, including neurodevelopmental impairments, cognitive deficits, delayed language and motor development, behavioral disorders, autism spectrum disorder, attention-deficit/hyperactivity disorder, and increased risk of later-life depression [1,41–43]. Early-life $PM_{2.5}$, exposure is also strongly linked to increased risks of childhood asthma [44]. Because prenatal and early are critical periods of rapid development, major contributors to ambient $PM_{2.5}$, such as coal combustion, can produce lasting health impacts that persist long after beyond exposure ends. In sum, this extensive body of evidence highlights the enduring and multifaceted public health burden posed by coal-related $PM_{2.5}$, even amid overall improvements in U.S. air quality. However, less is known about how these harms are established unevenly across geographically. In this study, we examine how the health effects of coal-related $PM_{2.5}$ vary across U.S. counties.

2.3. Recognitional and distributional justice of $PM_{2.5}$ exposure

Declines in coal use over the past decade have generated measurable public health benefits, including substantial reductions in premature mortality attributable to $PM_{2.5}$ exposure [26]. Yet, these gains have not been shared evenly, because exposure to $PM_{2.5}$ remains strongly stratified across places and populations. A recent study [45] found that reductions in direct mining exposure measured by miner labor hours are associated with higher life expectancy; This effect differ in Appalachian mining communities, where increases in coal production and coal mining employment are associated with decreases in life expectancy. From a justice perspective, this pattern reflects distributional inequality: the benefits of cleaner air and longer life expectancy accrue unevenly, while pollution burdens persist for some communities. Although coal-related emissions reductions have improved average population health and life expectancy [9], it remains unclear how equitably these improvements are distributed and how they intersect with broader social determinants of health [11]. This persistent unevenness also raises recognitional justice concerns, insofar as the lived experiences, vulnerabilities, and needs of historically marginalized groups are often insufficiently visible in air quality governance, health impact assessments, and policy evaluation. Therefore, these distributional and recognitional dynamics highlight the need for explicitly equity-focused analyses of air pollution and health.

A growing body of evidence demonstrates that racial and ethnic minority populations, particularly, Black communities, as well as low-income and socially vulnerable groups experience higher $PM_{2.5}$ exposures and greater associated health harms [7,41–43]. These patterns illustrate injustice in the allocation of environmental risks and health protections. In the U.S., county-level analyses reveal pronounced spatial inequalities: counties with higher income inequality and larger shares of racial and ethnic minority residents, especially Black residents, exhibit larger reductions in life expectancy attributable to air pollution [27]. Cross-national studies similarly indicate that income inequality strengthens the negative relationship between $PM_{2.5}$ exposure and life expectancy [28]. Health risks are further elevated among individuals with disabilities and those with compromised cardiovascular systems [32], underscoring recognitional justice issues when disability status and health vulnerability are not adequately accounted for in exposure mitigation, healthcare access, or risk communication. Consistent with these patterns, multiple studies find that Black populations in the U.S. bear a disproportionate burden of $PM_{2.5}$-related cardiovascular morbidity and mortality [46,47]. For example, a recent study [48] reports higher cerebrovascular mortality associated with $PM_{2.5}$ exposure, with the greatest risks concentrated in areas with larger proportions of Black residents, an inequity that persists even amid overall improvements and highlights the need to recognize and prioritize communities facing compounding structural vulnerabilities. To quantify how these inequities manifest across multiple health outcomes, we assessed regional disparities across nine physical and mental health indicators and evaluated whether improvements in air quality have produced equitable health gains.

2.4. Compounded risks of energy burdens and PM 2.5

EBs further amplify income-based disparities and health inequities [20]. From an energy justice perspective, EBs are not only a material constraint but also a distributional injustice, reflecting the unequal allocation of energy costs, thermal comfort, and related health protections across households and communities. EBs are disproportionately high among low-income and marginalized communities, which are often situated near power plants and industrial facilities and therefore face elevated pollution exposures [11]. These patterned co-occurrences of high EBs and high pollution also point to *recognitional injustice*, insofar as the lived realities, needs, and vulnerabilities of affected groups, particularly communities of color, low-income households, older adults, and people with disabilities, are often undervalued or insufficiently centered in energy, environmental, and public health decision-making.

Households with a high EB frequently rely on inefficient heating and cooling systems and pay more per square foot for energy, limiting their ability to maintain safe indoor temperatures and to access needed healthcare [18,20–22]. More than 25 million U.S. households report cutting back on food or medicine to cover energy bills, and millions live in homes kept at unsafe temperatures, conditions that can increase vulnerability to $PM_{2.5}$-related health risks [20]. These circumstances not only heighten environmental exposures but also constrain financial resources for healthcare and risk-reducing investments, thereby compounding health risks in structurally disadvantaged communities [11,49]. Even as national $PM_{2.5}$ concentrations have declined, improvements have been uneven; historically burdened communities remain disproportionately exposed, illustrating that aggregate progress can mask persistent distributional and recognitional inequities [7,11]. For example, a longitudinal analysis linking U.S. Census data (1970–2010) with county-level emissions by sector (transportation, agriculture, residential, commercial, industry, and energy) found persistent racial and socioeconomic disparities in exposure, with wealthier counties experiencing larger emissions reductions over time [7].

Despite this evidence, regional variation in the resulting health benefits remains poorly understood. While socioeconomic drivers of pollution exposure have been well documented [7], the relationship between $PM_{2.5}$ exposure and public health outcomes, especially how it varies across regions and interacts with social determinants of health, remains underexamined [11]. From a justice lens, this gap matters because *distributional benefits* (who gains from emissions reductions) and *recognitional dimensions* (whose risks and constraints are visible in policy design and evaluation) may vary systematically across regions. Additional reductions in $PM_{2.5}$ may therefore yield disproportionate health gains in particularly vulnerable regions [11]. Although prior studies have quantified the health benefits of reducing coal-fired power plant emissions [9], most emphasize national or state-level trends rather than regional or community-scale inequities. For instance, a recent study [26] assessed state-level changes in health impact estimates associated with

declining $PM_{2.5}$ from stationary fuel combustion sources, while another earlier research [10] examined the spatial distribution of co-benefits and costs associated with power-plant carbon standards. These studies demonstrate substantial health gains and offer important geographic detail. However, they do not directly test how the benefits of declining coal-related $PM_{2.5}$ differ across regions or how these benefits intersect with social determinants of health, including income, race/ethnicity, disability status, and access to healthcare, which are central to recognitional and distributional justice concerns [11].

Guided by the energy and environmental justice framework, this study contributes to the literature by finding regional disparities related to social determinants. We examine how coal-related $PM_{2.5}$ exposure and EBs jointly shape nine physical and mental health outcomes across 3108 U.S. counties across 49 U.S. mainland states. By integrating environmental and social determinants, we identify communities where health risks are compounded and provide evidence for targeted interventions to reduce both pollution exposure and social vulnerability. Whereas most prior work has examined pollution exposure and socioeconomic vulnerability separately, we evaluated their joint effects by testing whether household energy burden amplifies the health consequences of coal-related $PM_{2.5}$ exposure.

3. Methodology

3.1. Study area and time frame

This study's analysis covered 3108 counties in the 48 contiguous U.S. states plus the District of Columbia. We combined two temporal perspectives:

- 2016–2020: This study used annual coal $PM_{2.5}$ exposure estimates to describe recent trends and racial/ethnic disparities in exposure.
- 2020 cross-section: This study modeled county-level health outcomes, energy burden, and sociodemographic indicators for 2020, and link these to 2020 coal $PM_{2.5}$ exposure in spatial regression models.

Therefore, earlier years (2016–2019) were used to characterize the changing geography of coal $PM_{2.5}$, while the regression analysis focused on cross-sectional associations among the variables in 2020.

3.2. Dependent variables

The dependent variables (DVs) of the OLS regression model were county-level prevalences of nine health conditions among adults (≥18 years) in 2020, drawn from the US Centers for Disease Control and Prevention (CDC) Population Level Analysis and Community Estimates (PLACES) dataset [50]. Because direct collection of local, population-level health data is costly and often infeasible, the CDC Division of Population Health, working with the Robert Wood Johnson Foundation and the CDC Foundation, produced modeled small-area estimates to support public health surveillance and research. PlACES applied a small area estimation (SAE) approach called multilevel regression and post stratification (MRP) to generate prevalence estimates of 29 health measures (e.g., mental health, depression, cancer, chronic kidney disease, etc.) at four geographic levels (i.e., county, place, census tract, and ZCTA) national wide. In this study, we focused on nine health outcomes:

(1) *Poor mental health*: Percentage of adults reporting ≥14 days of poor mental health in the past 30 days.
(2) *Depression*: Percentage of adults ever told they have a depressive disorder.
(3) *Poor physical health*: Percentage of adults reporting ≥14 days of poor physical health in the past 30 days.
(4) *Coronary heart disease (CHD)*: Percentage ever told they have CHD or myocardial infarction.
(5) *COPD*: Percentage ever told they have COPD, emphysema, or chronic bronchitis.
(6) *Asthma*: Percentage ever told they currently have asthma.
(7) *Diabetes*: Percentage ever told they have diabetes (excluding gestational).
(8) *Stroke*: Percentage ever told they have had a stroke.
(9) *Insufficient sleep*: Percentage who report sleeping less than seven hours per 24-h period.

All health measures are modeled as continuous percentages at the county level.

3.3. Measures of IVs

3.3.1. Coal power plant-related fine particulate matter (coal $PM_{2.5}$)

The county-level coal $PM_{2.5}$ concentration data was sourced from the coal $PM_{2.5}$ exposure dataset generated by L. Henneman et al. [9]. This data was estimated using the HYSPIT Average Dispersion (HyADS) model, which employs monthly SO_2 emission information from 1237 coal electricity generation units operating since at least 1999. Technically, HyADS conducts simulations to track the air transport and dispersion of 100 air parcels at 6-h intervals each day for each coal generation unit over a multi-year horizon. The model derives monthly gridded air parcel counts for each power unit. By multiplying these gridded air parcel counts by each unit's monthly SO_2 emissions and averaging the monthly results, annual coal $PM_{2.5}$ concentrations were obtained. Since this estimation only accounts for a portion of total ambient $PM_{2.5}$ pollution (like contributions such as coal $PM_{2.5}$ from NO_x not included), the output of HyADS is regressed against coal $PM_{2.5}$ source impacts estimated with a more precise but complex model, the Hybrid Community Multiscale Air Quality model and Decoupled Direct Method (CMAQ-DDM). This regression ensures that HyADS estimations closely match the $PM_{2.5}$ impacts estimated by the Hybrid CMAQ-DDM model for the year 2005. Based on the power plant–specific $PM_{2.5}$ concentration estimation provided by the HyADS model, we can further calculate the population-weighted exposure (PWE) from each coal unit u in each year y for each demographic group g as:

$$PWE_{u,y,g} = \sum_{n=1}^{N} PM_{u,y,n} \times \left(\frac{P_{y,g,n}}{P_{y,g,total}} \right) \quad (1)$$

where $n = 1, 2, \ldots, N$ represents grid cell locations, $PM_{u,y,n}$ is the coal $PM_{2.5}$ concentration contributed by unit u in grid cell n in year y, $P_{y,g,n}$ is the population of group g in grid cell n in year y. $P_{y,g,total}$ is the total population of group g in grid cell n in year y.

3.3.2. Energy burdens (EBs)

EBs measure the percentage of a household's income spent on energy (including electricity, natural gas, etc.) bills, reflecting the financial strain energy costs place on a household's budget. A household with 6% or greater EB is considered as a high household EB [49]. The Low-income Energy Affordability Data (LEAD) tool [51] developed by the National Renewable Energy Laboratory (NREL) provides five-year data (2014–2018) on average annual energy costs and average EBs of the US at various levels—national, state, county, tribal, city, disadvantaged communities, and census tract boundaries. Since the LEAD tool only offers five-year average EB data from 2014 to 2018, we estimated the 2020 EB data using the LEAD data along with energy expenditure and income data from both 2020 and 2014–2018. Specifically, the household annual income data before taxes were sourced from the ACS [52]. This income data, combined with energy expenditure information, allowed us to estimate the 2020 county-level EBs for low-income households (LIHs) in the U.S. The historical data on energy expenditure (electricity, fuel, and natural gas) and EBs for LIHs were obtained from the LEAD tool by NREL and the U.S. Department of Energy (DOE), as well as from the Energy Information Administration (EIA). The 2020

county-level EB data were technically estimated based on the 2014–2018 county-level information on LIHs' annual income and 2020 state-level energy expenditure data, using the methodology outlined in reference [53] as:

$$EB_{c,2020} = EB_{c,2014-2018} \times \left(\frac{EE_{s,2020}}{EE_{s,2014-18}}\right) \times \left(\frac{INC_{c,2020}}{INC_{c,2014-18}}\right) \quad (2)$$

where $EE_{s,t}$ denotes average residential LIH energy expenditure for state s in period t, and $INC_{c,t}$ is median LIH income for county c. This approach preserves cross-county differences from the LEAD baseline while incorporating observed changes in income and energy prices at state level. The resulting county-level LIH EB in 2020 ($EB_{c,2020}$) is used as an independent variable (IV) in all regression models.

3.3.3. Sociodemographic variables

To capture social determinants of health and align with energy and environmental justice frameworks, we included the following county-level covariates:

(1) *Lack of health insurance*: Percentage of adults aged ≥18 years without health insurance in 2020, from CDC PLACES [50].
(2) *Minority status*: Percentage of the population identifying as non-white (all races and ethnicities other than non-Hispanic white), based on 2020 Census age, gender, and race data [52].
(3) *Women population*: Percentage of the population that is women [52].
(4) *Older adults*: Percentage of the adult population aged 65 years or older [52].
(5) *Disability*: Percentage of adults reporting any disability [52].

We selected these variables because prior work has shown that race/ethnicity, age, disability status, insurance coverage, and gender are strongly associated with both pollution exposure and health outcomes [7,32,41–43]. Together with EBs, they represent a set of structural factors central to energy-justice and public-health literatures. In this study, we focus on a single cross-section (2020) and thus do not explicitly model population change; sensitivity analyses including population density and population change did not materially alter the estimated effects of coal $PM_{2.5}$ and EBs.

3.4. Spatial dependence and bivariate hot spot analysis

Since health and pollution outcomes often exhibit spatial clustering, we first assessed spatial autocorrelation using Moran's I. For each health indicator and for coal $PM_{2.5}$, we computed Moran's I on county-level values using a queen contiguity spatial weights matrix (counties sharing an edge or vertex). Significant positive Moran's I values (e.g., 0.36–0.69 for OLS residuals, $p < 0.001$) indicate that counties with similar outcomes tend to be geographically clustered, violating the independence assumption of standard OLS regression.

To explore distribution patterns and environmental inequity, we conducted a bivariate hot spot analysis using the Getis-Ord Gi* statistic [54,55]. For each spatial unit (county) i, the G_i* statistic is:

$$G_i^* = \frac{\sum_j w_{ij} x_j}{\sum_j x_j} \quad (3)$$

where x_j represents the attribute value at location j, and w_{ij} is the spatial weight between locations i and j. A high positive G_i^* value indicates a hot spot, where high values cluster together, whereas a low negative G_i^* value indicates a cold spot, where low values are concentrated. The statistical significance of these clusters is determined using a permutation-based Monte Carlo approach, generating p-values to assess whether observed clustering deviates from a random spatial distribution.

We computed G_i^* separately for coal $PM_{2.5}$ and for the percentage of minority population, classify counties into hot spots, cold spots, or non-significant areas at 90%, 95%, and 99% confidence levels, and then overlay the two classifications. Counties that are simultaneous hot spots for both coal $PM_{2.5}$ and minority population are labeled bivariate hot spots, indicating locations where marginalized communities experience disproportionately high exposure to coal-related pollution. Analogous maps are produced for coal $PM_{2.5}$ and each health outcome.

3.5. Regression analysis

To quantify the associations between coal $PM_{2.5}$, EBs, sociodemographic factors, and health outcomes, we estimated four classes of models, i.e., OLS without state control variables; OLS with state control variables; spatial error model (SEM); and spatial lag model (SLM).

3.5.1. OLS regression model with and without the state control variables

As a baseline, we first estimated OLS No State models without state fixed effects. For each health outcome Y_c and each county c, we specify:

$$Y_c = \beta_0 + \beta_1(Coal\ PM_{2.5,c}) + \beta_2(EB_c) + \cdots + \beta_7(Disability_c) + \epsilon_c \quad (4)$$

where, $\beta_0, \cdots, \beta_7$ are regression coefficients, and ϵ_c is the error term. These models include the seven IVs described above and provide initial estimates of the direction and magnitude of associations between coal $PM_{2.5}$, EBs, and health outcomes.

To account for unobserved heterogeneity across states, such as differences in policies, health systems, or baseline disease burdens, we next estimated OLS models with state fixed effects. In this specification, we augmented the baseline model with a full set of state indicator variables D_{cs} and dropped the global intercept to avoid perfect collinearity, so that:

$$Y_c = \sum_s \alpha_s D_{cs} + \beta_1(Coal\ PM_{2.5,c}) + \beta_2(EB_c) + \cdots + \beta_7(Disability_c) + \epsilon_c \quad (5)$$

Including state fixed effects substantially improved model fit and reduced omitted-variable bias arising from unmeasured state-level factors.

3.5.2. SEM and SLM

Because residual diagnostics from the OLS models indicated spatial autocorrelation, we then estimated SEMs. In the SEM formulation, the disturbance term is modeled as:

$$\epsilon = \lambda W\varepsilon + u \quad (6)$$

where W is the queen-contiguity spatial weights matrix, λ is the spatial error coefficient, and u is an i.i.d. error term. This specification allows spatially structured unobserved factors, such as regional economic conditions or unmeasured environmental exposures, to be captured in the error process rather than biasing the estimated associations.

To explicitly represent spatial spillovers in health outcomes, we also fit spatial lag models (SLMs). In the SLM, the outcome vector Y is specified as:

$$Y = \rho WY + X\beta + \varepsilon \quad (7)$$

where WY is the spatially lagged DV (the weighted average of neighboring counties' outcomes), ρ measures the strength of spatial dependence, X is the matrix of covariates (including coal $PM_{2.5}$, EBs, and sociodemographic), and ε is an error term. This structure assumes that health outcomes in each county are influenced not only by local characteristics but also by conditions in neighboring counties, consistent with regional health dynamics.

All spatial analyses and mapping were conducted in Python (package *geopandas*, *spatialreg*). Regression diagnostics included assessment of multicollinearity (variance inflation factors), heteroskedasticity, and

spatial autocorrelation in residuals (Moran's I). Finally, to highlight regional differences, we partitioned counties into an Eastern and a Western region based on coal $PM_{2.5}$ patterns and conventional U.S. Census regions. We re-estimated spatial lag models separately for each region to examine how the association between coal $PM_{2.5}$ and health outcomes differs geographically.

4. Results and discussion

4.1. Results of OLS regression models

4.1.1. OLS regression model with and without state fix effects

We first compared OLS specifications estimated with and without state-level dummy variables (state fixed effects). In the baseline OLS models without state controls (hereafter, 'OLS No State," reported in Supplementary Tables S1-S8), we included seven independent variables (IVs) listed in Table 1. We estimated nine separate models, one for each health outcome. All baseline models were statistically significant ($p < 0.001$), with adjusted R^2 values ranging from 0.40 to 0.79 and AIC values between 2859 and 10,018. Next, we estimated OLS models that additionally included state dummy variables (hereafter, "OLS State," Supplementary Tables S1–S8). To avoid perfect multicollinearity (the "dummy variable trap"), we estimated these models without an intercept and included a dummy variable for each state. These models were also highly significant ($p < 0.001$) and exhibited substantially higher explanatory power, with adjusted R^2 values ranging from 0.85 to 0.92. Based on the diagnostic statistics, the inclusion of state dummies enhanced the model's fit and improved the estimation. Including state controls helps account for persistent, state-specific differences in population health that may reflect structural and policy contexts, such as health insurance coverage, income distributions, and demographic composition, that vary systematically across states; however, residual diagnostics from the baseline OLS models indicate remaining spatial dependence: Moran's I for the residuals was consistently large and positive (0.36–0.69; $p < 0.001$), suggesting substantial spatial autocorrelation and motivating spatial regression extensions in subsequent analyses.

4.1.2. Spatial error model

Conventional OLS estimation may underestimate standard errors when spatial dependence is present, because the assumption of independent errors is violated. To address this issue, we estimated a spatial error model (SEM), which augments the regression with a spatially correlated error component to capture unobserved influences that cluster across neighboring locations. We applied the SEM using the same set of IVs as in the OLS specifications. Overall, SEM results were consistent with the OLS findings, but model fit improved, as indicated by higher adjusted R^2 values and lower AIC. The spatial error parameter (λ) was large, positive, and statistically significant across outcomes ($\lambda = 0.78$–0.91), confirming that a substantial share of the unexplained variation in health outcomes was spatially structured (Supplementary Tables S1–S8).

However, Moran's I tests on SEM residuals remained significant ($p < 0.001$), suggesting that some spatial dependence persisted. This remaining autocorrelation may reflect omitted spatially clustered factors shared across neighboring counties. Importantly, the SEM attributes spatial dependence to unobserved influences in the error term and assumes these unobserved factors are not correlated with the included covariates.

4.1.3. Spatial lag model

To explicitly model geographic spillovers in county health outcomes, we estimated spatial lag models that incorporate dependence in the DVs across neighboring locations. Results from the LM–Lag diagnostics based on the baseline OLS specifications (Supplementary Table S10) indicated significant spatial dependence, motivating the use of a spatial lag (SAR) framework. In a spatial lag model, the outcome in county *i* was partly explained by surrounding counties *j* through a spatially lagged DV, *Wy*. Intuitively, *Wy* represents the weighted average of health outcomes in neighboring counties, capturing clustering and diffusion processes (e.g., shared healthcare markets, regional labor conditions, environmental exposures, and social dynamics). Specifically, the spatially lagged DV averaged the values of the DV from neighboring counties, helping to address autocorrelation in the model. For example, we expected mental health prevalence in each county to be related to mental health prevalence in adjacent counties.

Spatial relationships were defined using the queen contiguity criterion, in which counties sharing either a boundary or a vertex were treated as neighbors. We constructed a spatial weights matrix *W* based on this criterion and row-standardized it so that each county's neighbor weights sum to one. The spatial lag model included the same covariates as the standard OLS specification, with the addition of *Wy* (the spatial lag of the health outcome).

All spatial lag models were statistically significant and demonstrated improved fit relative to the standard OLS models, with log-likelihood values ranging from −4756.6 to −1232.2 and AIC values between 2482.4 and 9531.2. The estimated coefficients on the spatially lagged DV were positive and statistically significant across outcomes (including mental health, physical health, CHD, COPD, diabetes, stroke, and sleep), indicating meaningful inter-county spillovers in these health conditions. Consistent with this result, incorporating the spatial lag term improved model performance relative to both OLS and SEM specifications, as reflected in higher adjusted R^2 and stronger log-likelihood values.

Table 1
Coefficients and adjusted R^2 values of county-level spatial lag regression models for health outcomes in the U.S.

IV	Mental Health	De-pression	Physical Health	CHD	COPD	Asthma	Diabetes	Stroke	Sleeping < 7-h
Coal- PM2.5	0.02*	−0.01	0.08***	0.20***	0.13***	−0.01	0.34***	0.08***	0.12***
Health insurance	0.05***	0.002	0.23***	0.30***	0.12***	−0.11***	0.85***	0.15***	0.09***
Minority status	−0.01	−0.10***	0.04***	0.06***	0.002	0.14***	0.75***	0.28***	0.20***
Energy burden	0.11***	0.05***	0.15***	0.27***	0.14***	0.11***	0.37***	0.17***	0.05***
Women	0.09***	0.11***	0.03***	0.01	0.03***	0.14***	0.09***	0.02***	−0.03***
> Age 65	−0.37***	−0.19***	0.09***	0.81***	0.20***	−0.22***	0.77***	0.32***	−0.18***
Disability	0.22***	0.15***	0.30***	0.41***	0.27***	0.24***	0.60***	0.24***	0.20***
w-DV	0.69***	0.84***	0.49***	0.29***	0.56***	0.69***	0.31***	0.27***	0.61***
Adjust R^2	0.83	0.86	0.78	0.84	0.81	0.74	0.82	0.83	0.78

Note: 'w-DV' variable relates the values of a dependent variable (DV) at a given county to those in surrounding counties. In contrast to the coefficients of the other variables, which reflect global averages, variables such as mental health reflect the average impacts on the mental health of every county's neighboring county. A 1% increase in the mental health of surrounding counties correlates to a 0.69% increase in mental health for the average county, and this interpretation can be held for all other `w-` variables. IV = independent variable; w-DV – weighted dependent variables.

* $p < 0.05$.

*** $p < 0.001$.

4.2. Regional disparities in coal $PM_{2.5}$ exposure despite overall reductions

Our analysis indicates substantial but uneven reductions in coal-related $PM_{2.5}$ emissions across 49 U.S. states from 2016 to 2020. Annual state-level decreases ranged from 29% to 63%, with the largest declines observed in Montana (MT, 63.1%), Oklahoma (OK, 62.8%), Michigan (MI, 61.9%), and Indiana (IN, 59.0%). States with relatively high coal $PM_{2.5}$ concentrations in 2016, such as Kentucky, Pennsylvania, and Ohio (Fig. S1), also experienced marked reductions over the subsequent years. County-level patterns reveal even stronger spatial heterogeneity. In 2020, coal $PM_{2.5}$ concentrations remained highest across parts of the Southeast, Midwest, Mideast, and Northeast, while much lower levels were observed across much of the Western United States (Fig. 1). Changes in coal-related $PM_{2.5}$ at the county level between 2016 and 2020 varied widely, ranging from a 77.6% reduction to an 8.2% increase across the 3108 counties analyzed (Supplementary Tables S11–S12). Counties with the largest declines included Napa, California (77.6%), Freestone, Texas (77.5%), and Solano, California (77.0%). In contrast, several counties in Washington State showed little improvement or worsening conditions, including Grays Harbor (−8.1%), Mason (−2.3%), and Pacific (5.0%).

This study also highlights significant racial and ethnic disparities in coal-related $PM_{2.5}$ exposure in the U.S. [9]. At the national level, Black and White populations consistently experience the highest population-weighted coal-related $PM_{2.5}$ exposures (Fig. 2a). However, disaggregating patterns across the four Census regions, including South, West, Northeast, and Midwest (Fig. S2), reveals pronounced heterogeneity both within regions and across racial/ethnic groups (Fig. 2b). For instance, in the Western U.S., Native American, Pacific Islander, and Hispanic populations experience the highest exposure levels, with Native Americans facing coal $PM_{2.5}$ exposures nearly 1.8 times the national average (Fig. 2b). Additional regional contrasts were evident (Fig. 2c–f): the highest average exposures were observed among Black population in the South (Fig. 2c), Native Americans in the West (Fig. 2d), Asian population in the Northeast (Fig. 2e), and White population in the Midwest (Fig. 2f). Collectively, these results underscore that declines in coal $PM_{2.5}$ at the national scale have not translated into uniform improvements across demographic groups or regions.

To further examine these spatial patterns, we conducted a Hot Spot Analysis using the Getis-Ord Gi* statistic [55]. This approach identifies statistically significant spatial clusters of high or low values and is well suited for detecting geographic co-location of paired indicators. We mapped clusters where elevated coal $PM_{2.5}$ concentrations co-occur with higher proportions of minority residents at the county level (Fig. 3). Counties shown in red ("hot spots") represent locations where high coal $PM_{2.5}$ exposure spatially clusters with larger minority populations, whereas counties shown in blue ("cold spots") indicate clusters characterized by lower coal $PM_{2.5}$ exposure and smaller minority populations.

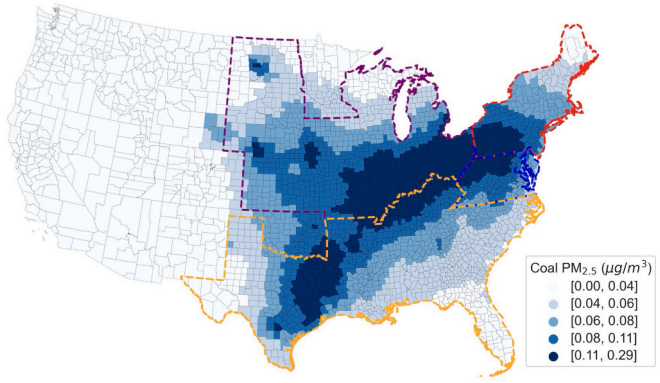


Fig. 1. Annual average coal $PM_{2.5}$ ($\mu g/m^3$) of counties in continental US in 2020. Darker blue colors indicate a higher coal $PM_{2.5}$ concentration level, while lighter blue colors indicate a lower coal $PM_{2.5}$ concentration level. The boundaries of U.S. regions with relatively high coal $PM_{2.5}$ concentration levels are highlighted as follows: the Midwest region in purple, the Northeast region in red, the Mideast region in blue, and the Southeast region in orange. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Fig. 4 illustrates the county-level prevalence of nine adult health outcomes in 2020 including mental health distress, depression, poor physical health, CHD, COPD, asthma, diabetes, stroke, and insufficient sleep. Hot spots for these outcomes are concentrated in the Southern United States and the Appalachian region (Fig. 4), revealing strong geographic clustering of adverse health burdens. Several outcomes exhibit closely aligned spatial patterns. For example, mental health distress (Fig. 4a) and depression (Fig. 4b) cluster in largely overlapping areas that also correspond to regions with elevated coal $PM_{2.5}$ exposure (Fig. 1), greater socioeconomic disadvantage (Fig. S4), and more limited healthcare access (Fig. S5).

We explored multivariate relationships between coal-$PM_{2.5}$ exposure, EBs and health using the OLS- based spatial regression models. Each model included seven predictors: (1) coal $PM_{2.5}$ concentration, (2) household EBs, and the county-level percentages of residents who are: (3) uninsured, (4) non-White, (5) women, (6) aged 65 or older, and (7) living with a disability. To address spatial autocorrelation, we estimated spatial lag specifications that include a spatially lagged DV, allowing health outcomes in each county to be related to outcomes in neighboring counties. The DVs were county-level adult prevalence rates for nine outcomes: poor self-reported mental health (≥14 days/month), diagnosed depression, poor physical health, CHD, COPD (including emphysema and chronic bronchitis), asthma, diabetes, stroke, and insufficient sleep (< 7 h/night).

Our spatial lag regression models for mental and physical health (Table 1) indicated that almost all IVs were statistically significant ($p < 0.05$), with adjusted R^2 values of 0.83 for mental health and 0.78 for physical health outcomes. A 1% increase in coal-related $PM_{2.5}$ exposure was associated with a 0.02% increase in the prevalence of poor mental health and a 0.08% increase in poor physical health. Additionally, CHD, COPD, diabetes, stroke, and insufficient sleep duration were significantly associated with coal $PM_{2.5}$ exposure. After controlling for other variables, a 1% increase in coal-related $PM_{2.5}$ predicted an increase of 0.08% to 0.34% in the prevalence of these five health conditions. Household EBs were significantly associated with all nine health outcomes: a 1% increase in EB predicted a 0.05% to 0.37% increase in prevalence across the nine health outcomes, with the strongest effects of CHD and diabetes observed for CHD and diabetes.

At the national level, our spatial regression models, which incorporated spatially lagged DVs, indicate statistically significant regional associations between coal $PM_{2.5}$ exposure and multiple health outcomes (Table 1). Because coal $PM_{2.5}$ concentrations were generally higher in eastern United States than in the West, we complemented the regression results with hot spot spatial to examine geographic clustering and regional variations (Fig. 3). The hot spot analysis identified clusters of counties, most prominently in the Southern and Southeastern United States, where high coal $PM_{2.5}$ concentrations co-occurred with elevated prevalence of the nine health outcomes. These hot spot counties (shown in red in Fig. 5) were especially apparent for mental health distress (Fig. 5a), depression (Fig. 5b), and poor physical health (Fig. 5c), and they were spatially contiguous with neighboring counties exhibiting similarly high values. In contrast, cold spots (shown in blue) were concentrated largely in the Western United States (Fig. 5). Similar hot–cold spot configurations were observed for CHD, COPD, asthma, diabetes, stroke, and insufficient sleep (Fig. 5d–i). In sum, these regional patterns indicate that counties with higher coal $PM_{2.5}$ exposure, particularly across the South and Southeast, also consistently experienced a greater burden of adverse mental and physical health outcomes.

Overall, coal-related $PM_{2.5}$ exposure in the eastern United States was associated with a broader and stronger set of adverse mental and

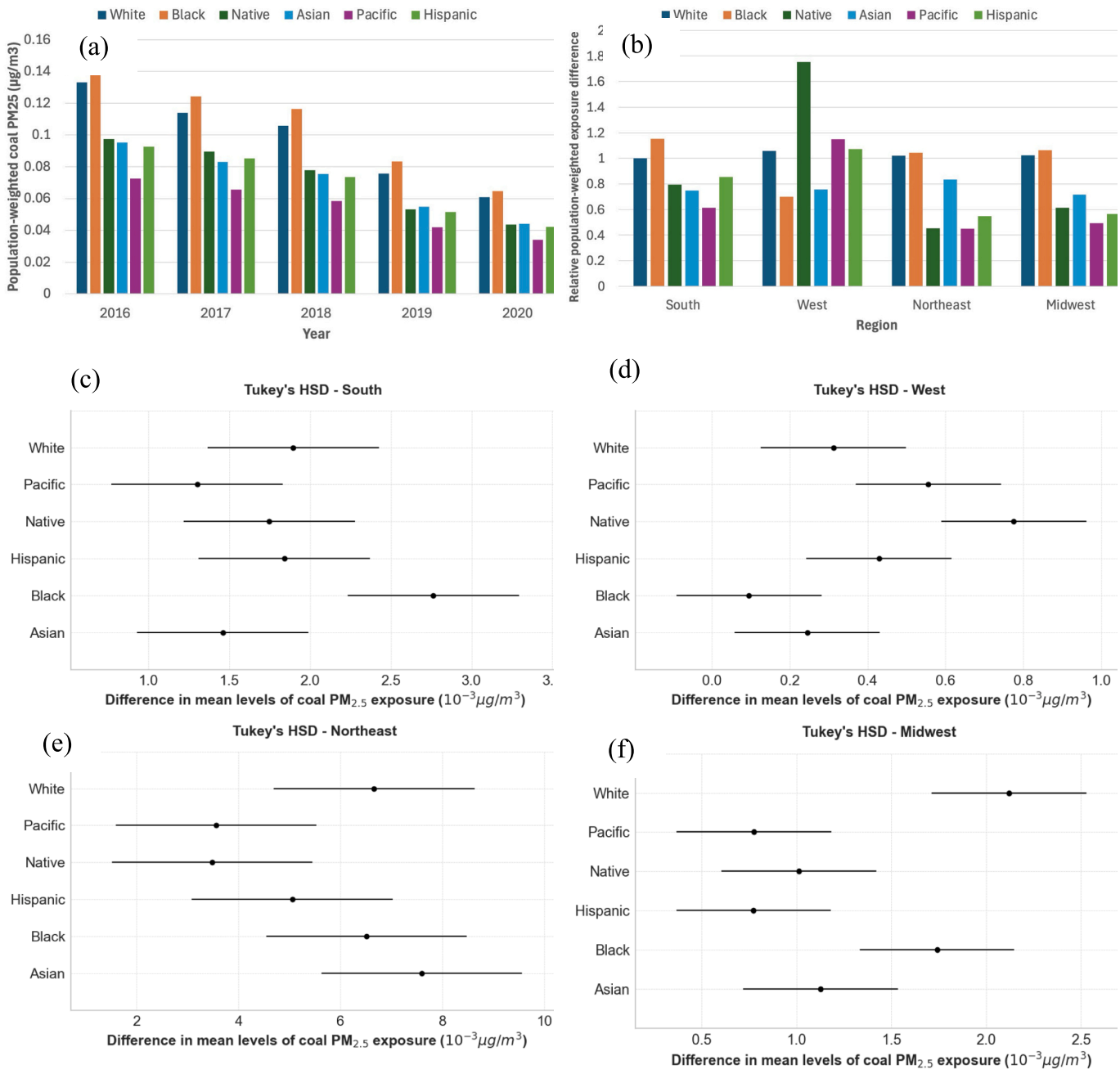


Fig. 2. Population-weighted $PM_{2.5}$ exposure from coal versus demographic groups in the contiguous United States. **(a)** National annual average exposure results from 2016 to 2020; **(b)** relative population-weighted exposure difference in 2020 for demographic groups defined in the 2020 US Census. **(c-f)** Tukey's simultaneous confidence intervals for one-way analysis of variance (ANOVA) of 2020 coal $PM_{2.5}$ exposure differences among racial groups in **(c)** South, **(d)** West, **(e)** Northeast and **(f)** Midwest regions of the US (see Fig. S2). Dots show estimated differences in the mean coal $PM_{2.5}$ levels between the groups, with 95% confidence intervals shown by horizontal bars.

physical health outcomes than in the West. To examine this regional heterogeneity, we divided states into an Eastern (Fig. 6a) and Western region (Fig. 6c) based on national coal $PM_{2.5}$ concentration patterns and then r e-estimated the same spatial lag regression models separately for each region using the previously defined independent and dependent variables.

In the Eastern region, a 1% increase in coal $PM_{2.5}$ was strongly associated ($p < 0.001$) with higher prevalence of poor mental health (0.07%) and poor physical health (0.13%), as well as increased prevalence of CHD (0.17%), COPD (0.13%), asthma (0.32%), diabetes (0.09%), stroke (0.30%), and insufficient sleep (0.09%) (Fig. 6b and Fig. 6d). In the Western region (Fig. 6c–d), coal $PM_{2.5}$ was associated with fewer outcomes. A 1% increase in coal $PM_{2.5}$ predicted statistically significant increases ($p < 0.05$) in depression (0.04%), CHD (0.20%), COPD (0.11%), diabetes (0.24%), stroke (0.10%), and sleep deprivation (0.08%), but it was not significantly associated with poor mental health, poor physical health, or asthma.

5. Discussion

This study contributes to the literature by examining regional disparities in health outcomes associated with coal-related $PM_{2.5}$ exposure, EB, and their intersection with key social determinants of health. Using a novel, publicly accessible dataset, we analyzed nine health indicators

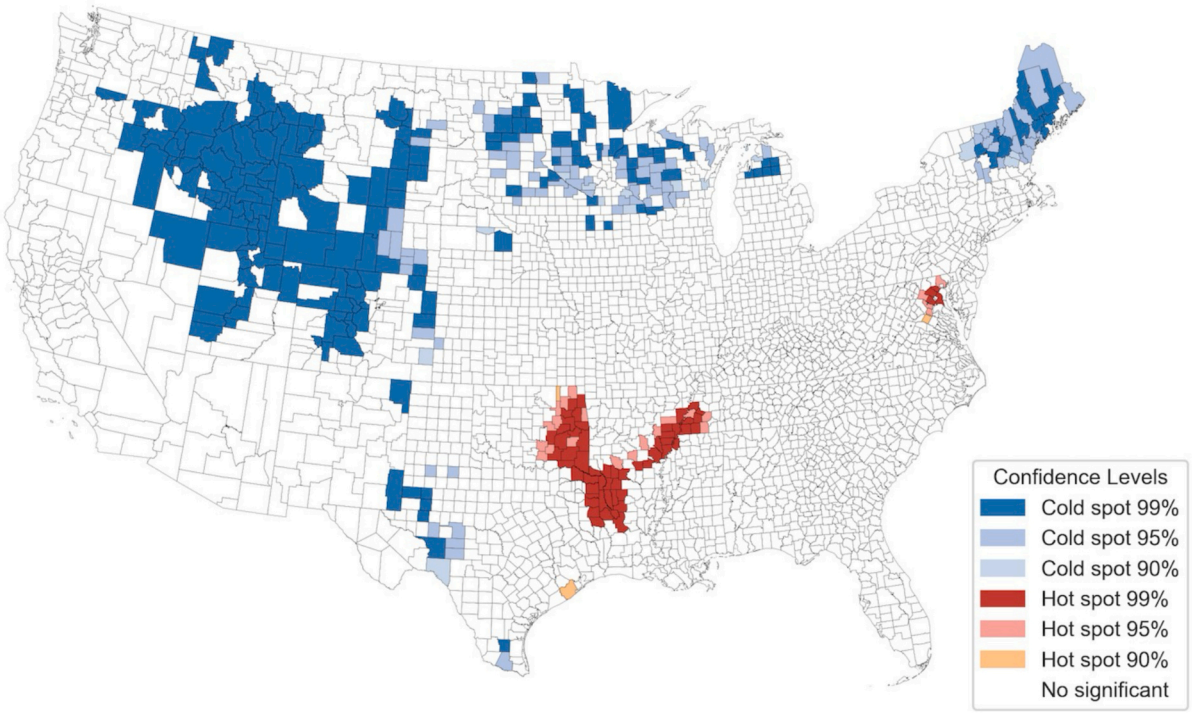


Fig. 3. Hot spot analysis of coal $PM_{2.5}$ and demographic groups. Minority population and coal $PM_{2.5}$ in 2016–2020. Deep red is the counties with a large minority population and high coal $PM_{2.5}$; lighter red is the counties with a large minority population and low coal $PM_{2.5}$; light blue is counties with a small minority population and high coal $PM_{2.5}$; deep blue is the counties that have a small minority population and low coal $PM_{2.5}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and identified pronounced geographic and socioeconomic patterning in long-term health burdens. Overall, our results align with prior evidence that fine particulate air pollution remains a major environmental health risk linked to premature mortality and chronic disease [2–4]. Beyond corroborating this broad relationship, our analysis offers three specific contributions: (1) identifying EB as a key vulnerability factor that amplifies pollution-related health risks; (2) documenting regional disparities in coal $PM_{2.5}$ exposure and associated health burdens; and (3) demonstrating the persistence of adverse health outcomes despite substantial national declines in coal $PM_{2.5}$.

Although coal-related $PM_{2.5}$ has declined markedly in the U.S. over the past decade, our findings indicate that coal-attributable health burdens persist, particularly for chronic conditions such as COPD, CHD, stroke, and diabetes. Because these outcomes accumulate over time, they may not fully reverse even after ambient concentrations decline, consistent with a temporal lag between exposure reductions and measurable changes in disease prevalence. Moreover, improvements in national averages do not necessarily translate into health gains for communities that continue to experience disproportionately high exposures. In the U.S. Southeast and Appalachia, regions where present-day coal $PM_{2.5}$ remains elevated [56–59], spatial heterogeneity and temporal lag suggest that disease burdens may persist among disadvantaged populations even as exposures decline, reflecting decades of sustained exposure. We interpret this pattern as "legacy risk," with three implications: (1) exposure reductions may yield delayed benefits for chronic diseases shaped by cumulative damage; (2) national decarbonization and air-quality gains can mask place-based inequities when remaining exposure is geographically concentrated; and (3) relying on national exposure trends alone may underestimate continuing health burdens in high-risk regions. Practically, these findings motivate regionally targeted strategies that pair emissions reductions with health-protective and resilience interventions in communities experiencing the highest historical and current coal $PM_{2.5}$ burdens.

A central contribution of this work is evidence that EB amplifies disparities in the health effects of coal-related $PM_{2.5}$, particularly for diabetes, CHD, stroke, and overall poor physical health. High-EB communities are more likely to be situated near coal-fired power plants and other pollution sources, compounding environmental and health risks [11]. They also often face housing and infrastructure constraints including older, energy-inefficient buildings and reliance on outdated heating and cooling systems that can increase indoor pollutant infiltration and exacerbate respiratory and cardiovascular stress [9,26]. Differences in EBs may therefore help explain why associations are stronger in the South and the broader Eastern United States, regions with a historical reliance on coal generation [7], than in many wealthier Western counties, where coal emissions have declined more sharply. Persistent inequities in infrastructure investment, regulatory protection, and resource allocation likely continue to heighten vulnerability to pollution-related health threats in marginalized communities [36,60,61].

Beyond physical health pathways, high EBs may also elevate vulnerability to cognitive and mental health harms by increasing chronic financial stress, disrupting sleep, and limiting access to preventive and mental healthcare services. When coupled with long-term $PM_{2.5}$ exposure, these constraints may plausibly intensify developmental and cognitive risks, particularly for children in households experiencing persistent energy insecurity [30,41–43].

Our findings also strengthen existing evidence linking coal-related $PM_{2.5}$ to higher prevalence of respiratory and cardiovascular conditions (e.g., COPD, CHD, stroke) as well as metabolic disorders such as diabetes [7,33]. One plausible mechanism is the disproportionate contribution of SO_2 emissions from coal plants to $PM_{2.5}$-related mortality [9]. We further observed associations between coal $PM_{2.5}$ and insufficient sleep, consistent with emerging research linking air pollution to sleep disruption [30]. Sleep disruption may function both as an outcome and as a pathway, given its role in cardiometabolic and mental health risk and its sensitivity to environmental exposures and household stressors.

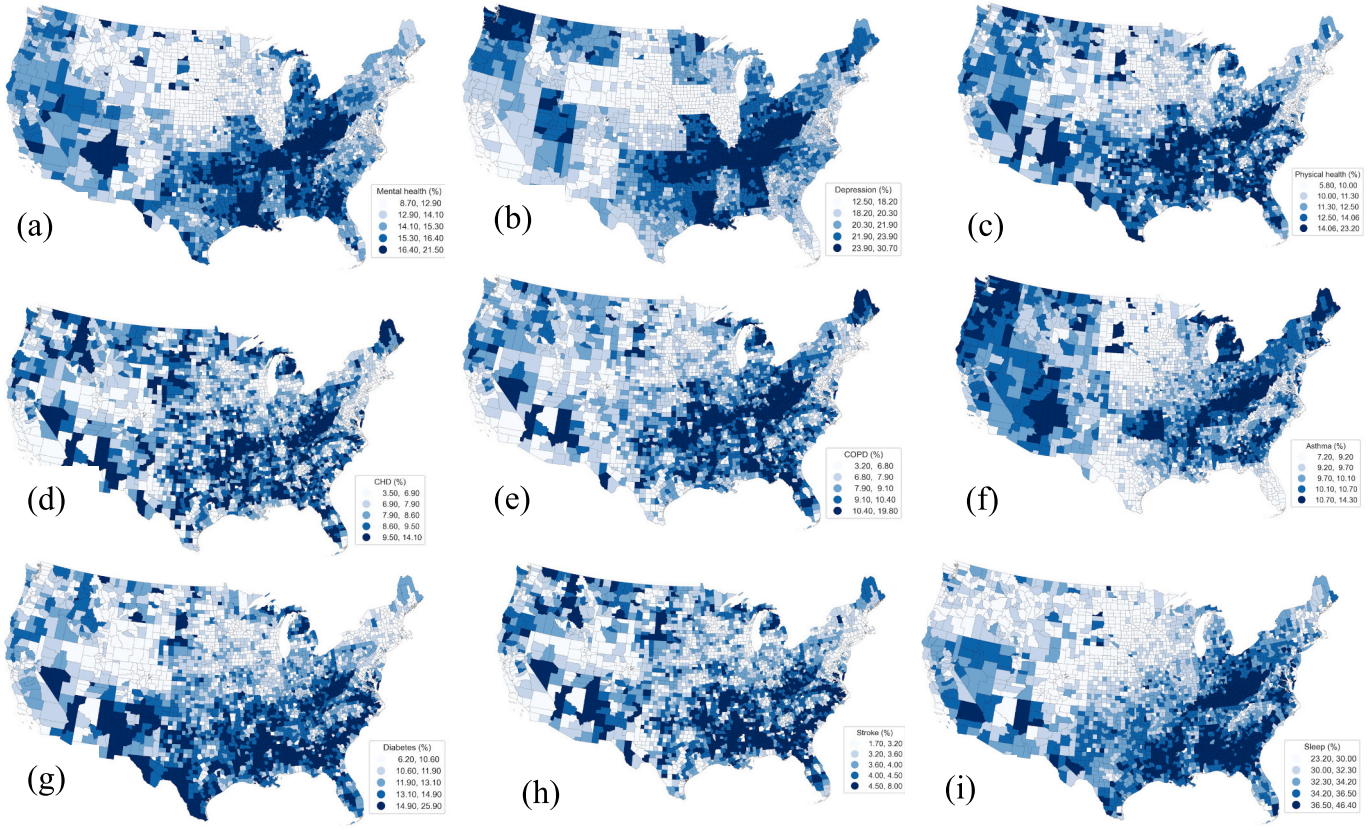


Fig. 4. County-level prevalence of 9 health measures of U.S. populations in 2020. The data are visualized using gradient shades of blue, where darker hues correspond to higher prevalence rates. (a) Mental health not good for ≥14 days among adults aged ≥18 years; (b) Depression among adults aged ≥18 years; (c) Physical health not good for ≥14 days among adults aged ≥18 years; (d) CHD among adults aged ≥18 years; (e) COPD among adults aged ≥18 years; (f) Asthma among adults aged ≥18 years; (g) Diagnosed diabetes among adults aged ≥18 years; (h) Stroke among adults aged ≥18 years; (i) Sleep less than 7 h among adults aged ≥18 years. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Importantly, we also found robust associations between coal-related $PM_{2.5}$ exposure and mental health outcomes, including depression and general mental distress, an increasingly recognized dimension of environmental health disparities [11,17]. This result is notable because mental health is often underrepresented in air-pollution burden assessments, yet it represents an increasingly recognized dimension of environmental health disparities. Chronic exposure to fine particulates has been linked to neuroinflammation, oxidative stress, and disruptions to neurotransmitter systems, which may contribute to cognitive impairment and mood disorders [30]. Prior studies also suggest that $PM_{2.5}$ exposure, including from coal combustion, may affect neurodevelopment and increase risks of disorders such as ADHD and autism spectrum disorder [41–43]. Taken together, our results underscore how ambient pollution and household energy insecurity jointly shape uneven health risks. In high-EB communities, pollution-related mental health harms may be compounded by chronic financial stress, thermal discomfort during heat or cold events, and limited access to mental health services, reinforcing the idea that environmental and social stressors can interact rather than merely add.

These findings have clear implications for policy and planning. First, they suggest that additional reductions in coal-related $PM_{2.5}$ are likely to yield the greatest health benefits when targeted to regions where high exposure and high EB intersect, such as parts of the Southeast and Appalachia. Second, they underscore the need for coordinated strategies that address both components of risk: upstream emissions mitigation (e.g., accelerated retirement of high-impact units or tighter controls where retirement is not immediate) and downstream vulnerability reduction (e.g., weatherization, HVAC modernization, indoor filtration, and energy affordability programs) in high-EB communities. Conceptually, this reflects an equity-centered approach in which reducing pollution is necessary but not sufficient unless households also have the resources and infrastructure to convert cleaner outdoor air into safer indoor environments where people spend most of their time.

Several limitations should be considered. First, the regression analysis is county-level and cross-sectional, which enables identification of robust associations and spatial patterning but does not support individual-level causal inference; ecological fallacy, residual confounding, and measurement error remain possible. Second, coal-related $PM_{2.5}$ is a source-specific exposure metric derived from a reduced-complexity atmospheric modeling framework (HyADS) calibrated to a more complex reference approach (Hybrid CMAQ-DDM) for a benchmark year. Although this approach yields unit-resolved exposure estimates well suited for national-scale equity analyses, it may not capture all pathways contributing to ambient $PM_{2.5}$ (e.g., chemistry beyond the SO_2-centered signal) and may not fully represent non-coal co-pollutant mixtures that can spatially co-vary with coal exposure. Third, county-level health prevalence may reflect geographic differences in diagnosis, reporting, healthcare access, and insurance coverage, factors that are themselves socially patterned. Finally, EB is a useful indicator of economic constraint, but it remains a proxy and cannot fully capture household coping strategies, indoor environmental conditions, housing quality, or housing-specific exposure pathways that may mediate health risks.

Future research should more closely examine how coal-related $PM_{2.5}$ interacts with socioeconomic vulnerability, especially in historically disadvantaged communities, using finer-grained local data. Longitudinal and quasi-experimental designs (e.g., assessing health changes following coal-unit retirements, emissions-control retrofits, or major policy shifts) would help determine whether recent exposure declines

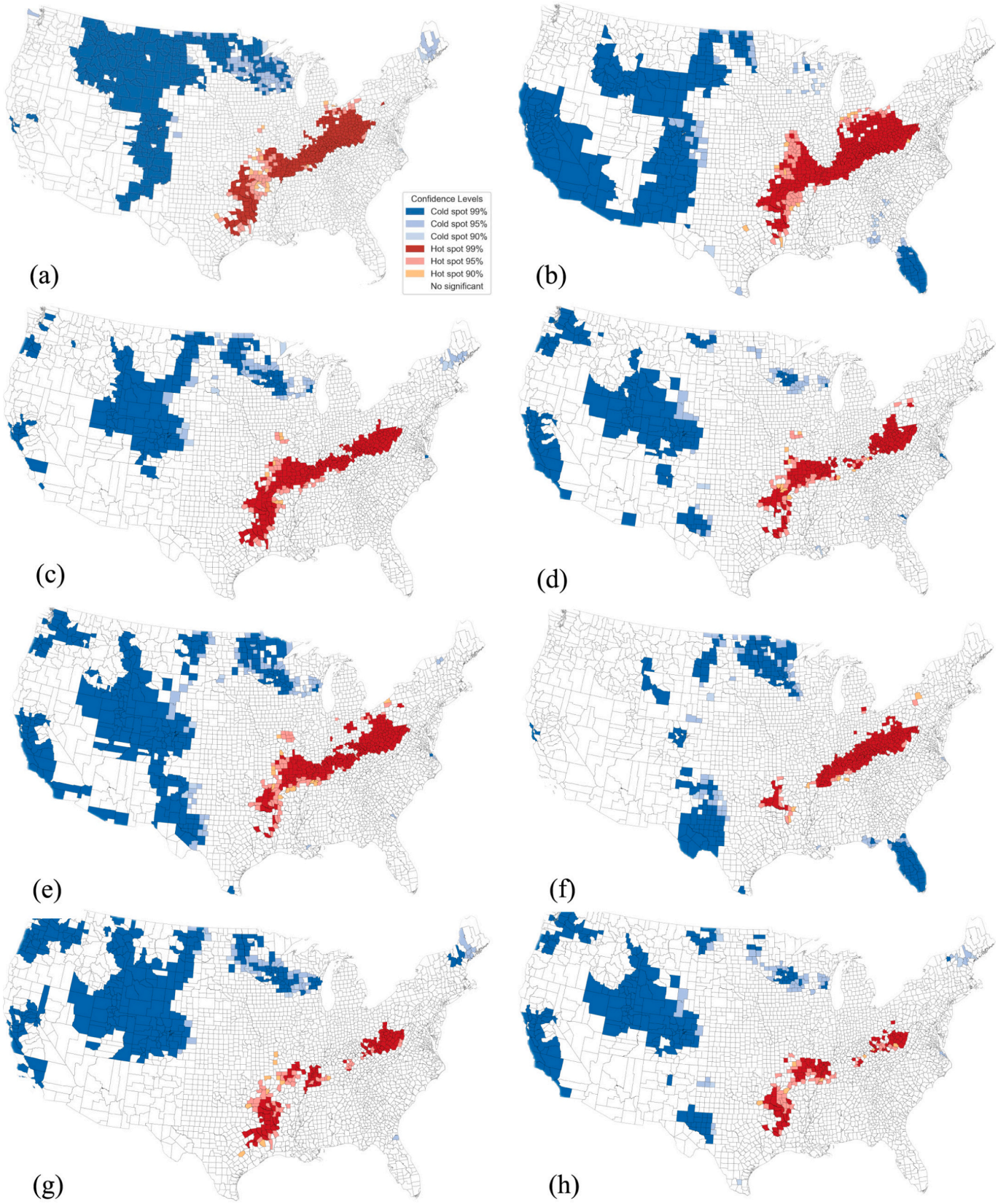


Fig. 5. Bivariate hot-spot maps highlighting hot and cold spots in spatial correlation between the nine mental/physical health outcomes in 2020 and coal $PM_{2.5}$ level in 2020. Red is the counties with high percentage of health issues and high coal $PM_{2.5}$; blue is the counties that have relatively low percentage of health issues and low coal $PM_{2.5}$. Each panel compares coal $PM_{2.5}$ versus a) mental health b) depression; c) physical health; d) CHD; e) COPD; f) asthma; g) diabetes; h) stroke; and i) sleeping less than 7-h per night. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

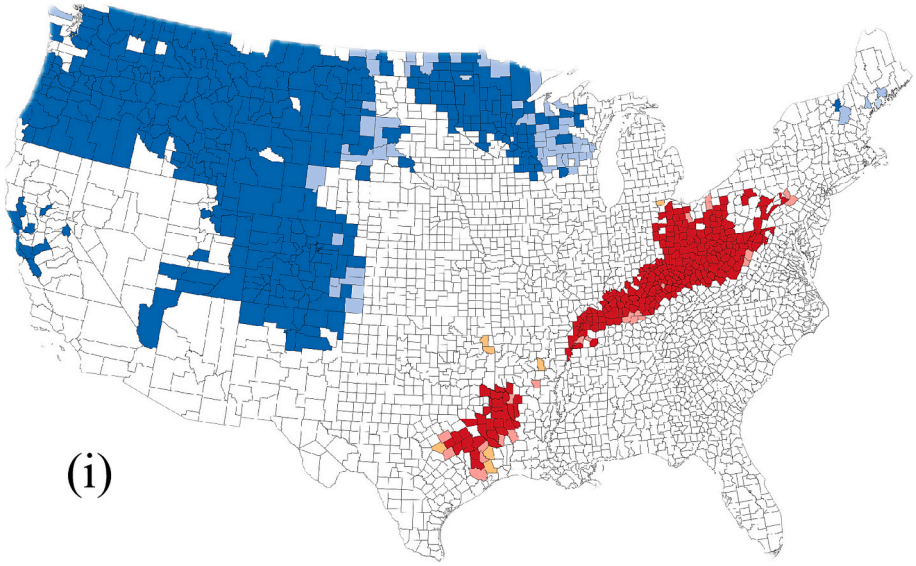


Fig. 5. (*continued*).

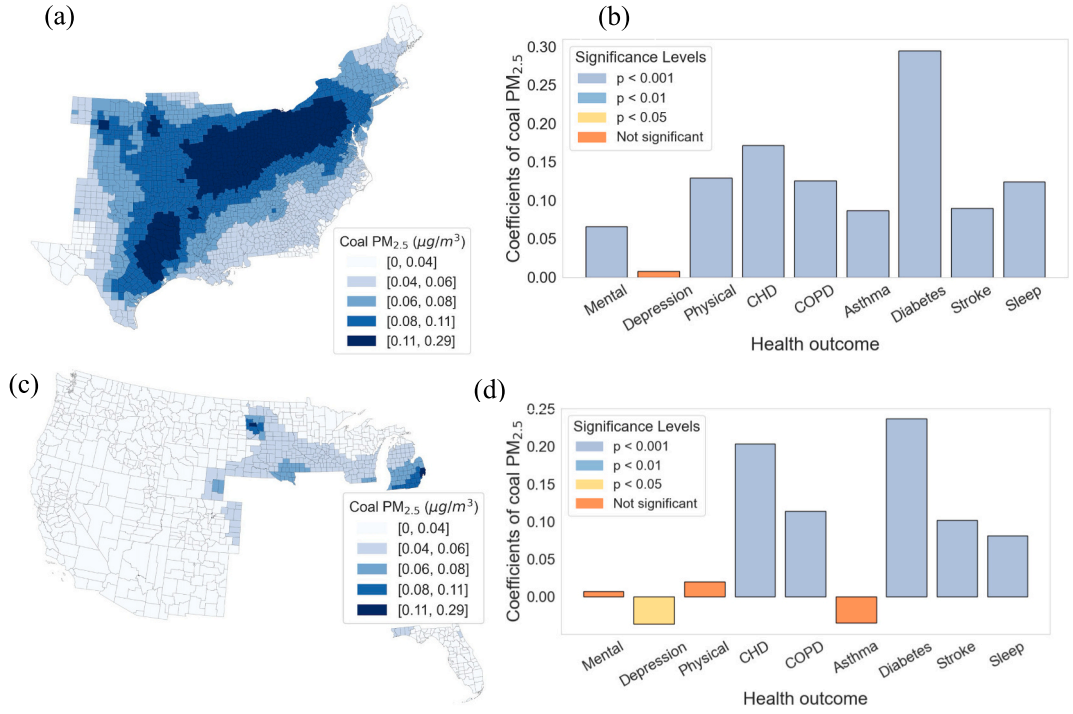


Fig. 6. Regional analysis results. (a) Counties within Eastern region and the corresponding coal $PM_{2.5}$ concentration level; (b) Coefficients of the coal $PM_{2.5}$ for the spatial lag and OLS models of Eastern region; (c) Counties within Western region and the corresponding coal $PM_{2.5}$ concentration level; (d) Coefficients of the coal $PM_{2.5}$ for the spatial lag OLS models of Western region.

translate into measurable improvements in physical and mental health, and whether these gains are larger, smaller, or delayed in high–EB areas. In addition, integrating coal-related $PM_{2.5}$ with other environmental stressors, such as transportation emissions, industrial chemical exposures, and extreme heat, would enable a more comprehensive assessment of cumulative risks. Expanding the analysis to multiple years of county-level health data would support evaluation of long-term trends and strengthen causal inference. Finally, incorporating race- and ethnicity-specific health outcomes would more directly illuminate inequities associated with coal pollution and better inform targeted, equity-centered policy interventions.

6. Conclusion

Although regulatory initiatives such as the Clean Air Act have substantially reduced pollution nationwide, coal-related $PM_{2.5}$ continues to pose persistent public health risks. This study offers three key insights. First, coal-related $PM_{2.5}$ remains measurably associated with adverse health outcomes in some regions even at today's lower concentrations, indicating that declining exposures do not necessarily translate into declining risk. Second, coal-related $PM_{2.5}$ burdens are disproportionately concentrated in the Southern and Eastern United States, suggesting that national analysis can mask ongoing, place-based vulnerability.

Third, EB emerges as a critical social vulnerability that amplifies pollution-related harms, shaping who benefits from environmental improvements and who remains at risk. Together, these findings improved average air quality does not guarantee equitable health gains, particularly for communities facing long-standing exposures and limited capacity to mitigate risk.

This study's results offer important policy implications. Persistent health burdens reflect "legacy risk" arising from cumulative exposure histories and lagged disease processes, while pronounced spatial heterogeneity underscores the need for targeted emission reductions and stronger monitoring in overburdened areas. Equally important, environmental policy must address the social determinants that shape vulnerability. High EB is closely linked to older, inefficient housing, greater susceptibility to exposure, and constrained access to resources that support health protection. Interventions such as weatherization, HVAC upgrades, improved filtration, and energy affordability support, should be treated as complementary public health strategies, not solely economic assistance. Pairing emissions reductions with investments that improve housing conditions and access to healthcare in high-burden communities is likely to yield larger and more equitable health gains than either approach alone.

Overall, this study demonstrates that coal-related $PM_{2.5}$ exposure continues to impose substantial health risks, particularly in regions where burdens remain concentrated. Reducing these inequities will require integrated, equity-centered strategies that both limit remaining emissions and strengthen community resilience by lowering energy insecurity and improving the social and environmental conditions that shape exposure and long-term health.

CRediT authorship contribution statement

Chien-fei Chen: Writing – review & editing, Writing – original draft, Conceptualization. **Hang Shuai:** Writing – review & editing, Writing – original draft, Formal analysis. **R. Alexander Bentley:** Writing – review & editing, Writing – original draft. **Lyudmyla Tsykalova:** Writing – review & editing, Writing – original draft. **Zhenglai Shen:** Writing – review & editing, Writing – original draft.

Inclusion and diversity

One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in social science. One or more of the authors of this paper self-identifies as received support from a program designed to increase minority representation in science.

Materials availability

No materials were used in this study.

Declaration of competing interest

The authors have no financial or non-financial interests associated with the material in this manuscript.

Acknowledgments

Funding: C.-F. Chen, H. Shuai and L. Tsykalova are supported by the Wellcome Thrust Foundation under G7 Climate and Mitation Policy Solution Award #. 227151/Z/23/Z.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.erss.2026.104631.

Data availability

This study used four nationally representative data sources, including the U.S. CDC PLACES data, coal PM2.5 data from the Open Science Framework [39], the U.S. Department of Energy's Low-Income Energy Affordability Data (LEAD) tool [34], and the American Community Survey (ACS). Codes are available upon request. Code requests should be directed to chienfc@clemson.edu.

References

[1] World Health Organization, Ambient (outdoor) air pollution. https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health, 2024 (accessed November 18, 2025).
[2] S. Lelieveld, J. Wilson, E. Dovrou, A. Mishra, P.S.J. Lakey, M. Shiraiwa, U. Poschl, T. Berkemeier, Hydroxyl radical production by air pollutants in epithelial lining fluid governed by interconversion and scavenging of reactive oxygen species, Environ. Sci. Technol. 55 (2021) 14069–14079, https://doi.org/10.1021/acs.est.1c03875.
[3] E.E. McDuffie, R.V. Martin, J.V. Spadaro, R. Burnett, S.J. Smith, P. O'Rourke, M. Brauer, Source sector and fuel contributions to ambient PM2.5 and attributable mortality across multiple spatial scales, Nat. Commun. 12 (2021) 3594, https://doi.org/10.1038/s41467-021-23853-y.
[4] Q. Wang, S. Liu, The effects and pathogenesis of PM2.5 and its components on chronic obstructive pulmonary disease, in: Int. J. Chron. Obstruct. Pulmon. Dis., 2023, pp. 493–506, https://doi.org/10.2147/COPD.S402122.
[5] K. Jenkins, Setting energy justice apart from the crowd: lessons from environmental and climate justice, Energy Res. Soc. Sci. 39 (2018) 117–121, https://doi.org/10.1016/j.erss.2017.11.015.
[6] U.S. Energy Information Administration, U.S. Energy Facts Explained. https://www.eia.gov/energyexplained/us-energy-facts/data-and-statistics.php, 2024 (accessed October 12, 2025).
[7] Y. Nunez, J. Benavides, J.A. Shearston, E.M. Krieger, M. Daouda, L.R. Henneman, M.A. Kioumourtzoglou, An environmental justice analysis of air pollution emissions in the United States from 1970 to 2010, Nat. Commun. 15 (2024) 268, https://doi.org/10.1038/s41467-023-43492-.
[8] C.A.I.I.I. Pope, A. Bhatnagar, J.P. McCracken, W. Abplanalp, D.J. Conklin, T. O'Toole, Exposure to fine particulate air pollution is associated with endothelial injury and systemic inflammation, Circ. Res. 119 (2016) 1204–1214, https://doi.org/10.1161/CIRCRESAHA.116.3092.
[9] L.R.F. Henneman, M.M. Rasel, C. Choirat, S.C. Anenberg, C. Zigler, Inequitable exposures to U.S. coal power plant-related PM2.5: 22 years and counting, Environ. Health Perspect. 131 (2023) 37005, https://doi.org/10.1289/EHP11605.
[10] J.J. Buonocore, K.F. Lambert, D. Burtraw, S. Sekar, C.T. Driscoll, An analysis of costs and health co-benefits for a US power plant carbon standard, PLoS One 11 (2016) e0156308, https://doi.org/10.1371/journal.pone.0158792.
[11] T.G. Reames, D.M. Daley, J.C. Pierce, Exploring the nexus of energy burden, social capital, and environmental quality in shaping health in US counties, Int. J. Environ. Res. Public Health 18 (2021) 1–13, https://doi.org/10.3390/ijerph18020620.
[12] P.A. Ayinla, L. Tsykalova, C.F. Chen, Regional dynamics of public attitudes toward climate change policy, regulatory strategies, and the energy transition in the United States, Energy Res. Soc. Sci. Soc. Sci. 127 (2025) 104291, https://doi.org/10.1016/j.erss.2025.104291.
[13] B.K. Sovacool, M.H. Dworkin, Energy justice: conceptual insights and practical applications, Appl. Energy 142 (2015) 435–444, https://doi.org/10.1016/j.apenergy.2015.01.002.
[14] R.J. Heffron, D. McCauley, The concept of energy justice across the disciplines, Energy Policy 105 (2017) 658–667, https://doi.org/10.1016/j.enpol.2017.03.018.
[15] S. Bouzarovski, N. Simcock, Spatializing energy justice, Energy Policy 107 (2017) 640–648, https://doi.org/10.1016/j.enpol.2017.03.064.
[16] T. Clemens, S. Turner, C. Dibben, Maternal exposure to ambient air pollution and fetal growth in north-East Scotland, Environ. Int. 107 (2017) 216–226, https://doi.org/10.1016/j.envint.2017.07.018.
[17] S. Roberts, L. Arseneault, B. Barratt, S. Beevers, A. Danese, C.L. Odgers, H.L. Fisher, Exploration of NO2 and PM2.5 air pollution and mental health problems, Psychiatry Res. 272 (2019) 8–17, https://doi.org/10.1016/j.psychres.2018.12.050.
[18] Z. Shen, C.-F. Chen, H. Zhou, N. Fefferman, S. Shrestha, Community vulnerability is the key determinant of diverse energy burdens, Energy Res. Soc. Sci. 97 (2023) 102949, https://doi.org/10.1016/j.erss.2022.102949.
[19] K. Jenkins, B.K. Sovacool, N. Mouter, N. Hacking, M.K. Burns, D. McCauley, The methodologies, geographies, and technologies of energy justice, Environ. Res. Lett. 16 (2021) 43009, https://doi.org/10.1088/1748-9326/abd78c.
[20] D. Hernández, Understanding 'energy insecurity' and why it matters to health, Soc. Sci. Med. 167 (2016) 1–10, https://doi.org/10.1016/j.socscimed.2016.08.029.
[21] C.F. Chen, W.A. Chen, J. Greig, C. Robinson, When concentrated disadvantage happens: exploring the nexus of energy and internet insecurities among vulnerable households, iScience 27 (2024) 109895, https://doi.org/10.1016/j.isci.2024.109895.
[22] C. Chen, J. Greig, H. Nelson, F. Li, When disadvantage collides: the concentrated disadvantage effects on energy insecurity and internet burdens in the U.S, Energy Res. Soc. Sci. 91 (2022) 102713, https://doi.org/10.1016/j.erss.2022.102713.

[23] D. McCauley, R. Heffron, Just transition: integrating climate, energy and environmental justice, Energy Policy 119 (2018) 1–7, https://doi.org/10.1016/j.enpol.2018.04.014.
[24] P. Mohai, D. Pellow, J.T. Roberts, Environmental justice, Annu. Rev. Environ. Resour. 34 (2009) 405–430, https://doi.org/10.1146/annurev-environ-082508-094348.
[25] C. Cannon, J. Bonnel, M. Padilla, D. Sulca, Along the energy justice continuum: an examination of waste, disposal, and rural communities, Energy Res. Soc. Sci. 96 (2023) 102948, https://doi.org/10.1016/j.erss.2023.102948.
[26] J.J. Buonocore, P. Salimifard, D.R. Michanowicz, J.G. Allen, A decade of the {US} energy mix transitioning away from coal, Environ. Res. Lett. 16 (2021) 54030, https://doi.org/10.1088/1748-9326/abe74c.
[27] A.K. Jorgenson, T.D. Hill, B. Clark, R.P. Thombs, P. Ore, K.S. Balistreri, J.E. Givens, Power, proximity, and physiology: does income inequality and racial composition amplify the impacts of air pollution on life expectancy in the United States? Environ. Res. Lett. 15 (2020) 24013, https://doi.org/10.1088/1748-9326/ab6789.
[28] A.K. Jorgenson, R.P. Thombs, B. Clark, J.E. Givens, T.D. Hill, X. Huang, O.M. Kelly, J.B. Fitzgerald, Inequality amplifies the negative association between life expectancy and air pollution: a cross-national longitudinal study, Sci. Total Environ. 758 (2021) 143705, https://doi.org/10.1016/j.scitotenv.2020.143705.
[29] U.S. Environmental Protection Agency, Summary of the Clean Air Act. https://www.epa.gov/laws-regulations/summary-clean-air-act, 2025 (accessed October 12, 2025).
[30] C.H. Zhang, L. Sears, J.V. Myers, G.N. Brock, C.G. Sears, K.M. Zierold, Proximity to coal-fired power plants and neurobehavioral symptoms in children, J. Expo. Sci. Environ. Epidemiol. 32 (2022) 124–134, https://doi.org/10.1038/s41370-021-00369-7.
[31] S. Chowdhury, A. Pozzer, A. Haines, K. Klingmuller, T. Munzel, P. Paasonen, A. Sharma, C. Venkataraman, J. Lelieveld, Global health burden of ambient PM2.5 and the contribution of anthropogenic black carbon and organic aerosols, Environ. Int. 159 (2022) 107020, https://doi.org/10.1016/j.envint.2021.107020.
[32] S. Basith, B. Manavalan, T.H. Shin, C.B. Park, W.S. Lee, J. Kim, G. Lee, The impact of fine particulate matter 2.5 on the cardiovascular system: a review of the invisible killer, Nanomaterials 12 (2022) 2656, https://doi.org/10.3390/nano12152656.
[33] D.E. Schraufnagel, The health effects of ultrafine particles, Exp. Mol. Med. 52 (2020) 311–317, https://doi.org/10.1038/s12276-020-0403-3.
[34] Y. Sethi, S. Mehta, I. Padda, P. Marlecha, A. Moinuddin, Impact of PM2.5 exposure on cardiovascular diseases [IPEC study]: an updated umbrella review of systematic reviews and meta-analyses, Eur. J. Prev. Cardiol. (2026), https://doi.org/10.1093/eurjpc/zwag005.
[35] G. Grande, J. Wu, P.L.S. Ljungman, M. Stafoggia, T. Bellander, D. Rizzuto, Long-term exposure to PM2.5 and cognitive decline: a longitudinal population-based study, J. Alzheimer's Dis 80 (2021) 591–599, https://doi.org/10.3233/JAD-200852.
[36] J. Liu, L.P. Clark, M.J. Bechle, A. Hajat, S.-Y. Kim, A.L. Robinson, L. Sheppard, A. A. Szpiro, J.D. Marshall, Disparities in air pollution exposure in the United States by race/ethnicity and income, 1990-2010, Environ. Health Perspect. 129 (2021) 127005, https://doi.org/10.1289/EHP8584.
[37] M. Davoudi, F. Barjasteh-Askari, H. Amini, et al., Association of suicide with short-term exposure to air pollution at different lag times: a systematic review and meta-analysis, Sci. Total Environ. 771 (2021) 144882, https://doi.org/10.1016/j.scitotenv.2020.144882.
[38] S. Heo, W. Lee, M.L. Bell, Suicide and associations with air pollution and ambient temperature: a systematic review and Meta-analysis, Int. J. Environ. Res. Public Health 18 (2021) 7699, https://doi.org/10.3390/ijerph18147699.
[39] J.D. King, S. Zhang, A. Cohen, Air pollution and mental health: associations, mechanisms and methods, Curr. Opin. Psychiatry 35 (2022) 192–199, https://doi.org/10.1097/YCO.0000000000000771.
[40] P.-C. Huang, J.E. Whang, M.-S. Lee, H.-S. Chang, J.-Y. Chen, Y.-G. Tsai, The effects of fine particulate matter on pediatric airway inflammation and lung function: an epidemiological perspective, Pediatr. Respirol. Crit. Care Med. 9 (2025) 69–75, https://doi.org/10.4103/prcm.prcm_7_25.
[41] E. Harris, Phasing out fossil fuels could reduce millions of excess deaths, JAMA 331 (2023) 16, https://doi.org/10.1001/jama.2023.24687.
[42] F. Perera, T.Y. Li, Z.J. Zhou, T. Yuan, Y.H. Chen, L. Qu, D. Tang, Benefits of reducing prenatal exposure to coal-burning pollutants to children's neurodevelopment in China, Environ. Health Perspect. 116 (2008) 1396–1400, https://doi.org/10.1289/ehp.11480.
[43] L. Sears, M.J. Smith, L. Cai, K.M. Zierold, The association of manganese overexposure and neurobehavioral function is moderated by arsenic: a metal mixture analysis of children living near coal ash storage sites, Neurotoxicology 103 (2024) 78–86, https://doi.org/10.1016/j.neuro.2024.06.005.
[44] G.D. Thurston, J.R. Balmes, E. Garcia, F.D. Gilliland, M.B. Rice, T. Schikowski, L.S. V. Winkle, I. Annesi-Maesano, E.G. Burchard, C. Carlsten, J.R. Harkema, H. Khreis, S.R. Kleeberger, U.P. Kodavanti, S.J. London, R. McConnell, D.B. Peden, K. E. Pinkerton, J. Reibman, C.W. White, Outdoor air pollution and new-onset airway disease, Ann. Am. Thorac. Soc. 17 (2020) 387–398.
[45] R.P. Thombs, B.K. Sovacool, A.K. Jorgenson, J.J. Buonocore, M.D. Willis, A not-so-just transition? Examining the effects of coal sector decline on life expectancy in US counties, Rural. Sociol. 91 (2026), https://doi.org/10.1111/ruso.70034.
[46] Y. Ma, E. Zang, I. Opara, Y. Lu, H.M. Krumholz, K. Chen, Racial/ethnic disparities in PM2.5-attributable cardiovascular mortality burden in the United States, Nat. Hum. Behav. 7 (2023) 2074–2083, https://doi.org/10.1038/s41562-023-01694-7.
[47] A.M. Weaver, L. McGuinn, L. Neas, J. Mirowsky, R.B. Devlin, R. Dhingra, C. Ward-Caviness, W.E. Cascio, W.E. Kraus, E.R. Hauser, Q. Di, J. Schwartz, D. Diaz-Sanchez, Neighborhood sociodemographic effects on the associations between long-term PM2.5 exposure and cardiovascular outcomes and diabetes mellitus, Environ. Epidemiol. 3 (2019) e038, https://doi.org/10.1097/EE9.0000000000000038.
[48] M. Yitshak-Sade, R. Nethery, Y. Abu Awad, F. Mealli, F. Dominici, I. Kloog, A. Zanobetti, Lowering air pollution levels in Massachusetts may prevent cardiovascular hospital admissions, J. Am. Coll. Cardiol. 75 (2020) 2642–2644, https://doi.org/10.1016/j.jacc.2020.03.056.
[49] A. Drehobl, L. Ross, R. Ayala, How High Are Household Energy Burdens?: An Assessment of National and Metropolitan Energy Burden across the United States. https://www.aceee.org/research-report/u2006, 2020.
[50] Centers for Disease Control and Prevention, PLACES: Local Data for Better Health. https://www.cdc.gov/places/index.html, 2023 (accessed November 18, 2025).
[51] U.S. Department of Energy, Low-income Energy Affordability Data (LEAD) Tool, 2024.
[52] United States Census Bureau, American Community Survey (ACS) Data. https://data.census.gov, 2022 (accessed September 14, 2025).
[53] C.F. Chen, J. Feng, N. Luke, C.P. Kuo, J.S. Fu, Localized energy burden, concentrated disadvantage, and the feminization of energy poverty, iScience 25 (2022) 104139, https://doi.org/10.1016/j.isci.2022.104139.
[54] J.K. Ord, A. Getis, Local spatial autocorrelation statistics: distributional issues and an application, Geogr. Anal. 27 (1995) 286–306, https://doi.org/10.1111/j.1538-4632.1995.tb00912.x.
[55] A. Getis, J.K. Ord, The analysis of spatial association by use of distance statistics, Geogr. Anal. 24 (1992) 189–206, https://doi.org/10.1111/j.1538-4632.1992.tb00261.x.
[56] L.P. Clark, D.B. Millet, J.D. Marshall, Changes in transportation-related air pollution exposures by race-ethnicity and socioeconomic status: outdoor nitrogen dioxide in the United States in 2000 and 2010, Environ. Health Perspect. 125 (2017) 97012, https://doi.org/10.1289/EHP959.
[57] L.P. Clark, D.B. Millet, J.D. Marshall, National patterns in environmental injustice and inequality: outdoor NO2 air pollution in the United States, PLoS One 9 (2014) e94431, https://doi.org/10.1371/journal.pone.0094431.
[58] I. Mikati, A.F. Benson, T.J. Luben, J.D. Sacks, J. Richmond-Bryant, Disparities in distribution of particulate matter emission sources by race and poverty status, Am. J. Public Health 108 (2018) 480–485, https://doi.org/10.2105/AJPH.2017.304297.
[59] M.L. Miranda, S.E. Edwards, M.H. Keating, C.J. Paul, Making the environmental justice grade: the relative burden of air pollution exposure in the United States, Int. J. Environ. Res. Public Health 8 (2011) 1755–1771, https://doi.org/10.3390/ijerph8061755.
[60] A. Jbaily, X. Zhou, J. Liu, T.-H. Lee, L. Kamareddine, S. Verguet, F. Dominici, Air pollution exposure disparities across US population and income groups, Nature 601 (2022) 228–233, https://doi.org/10.1038/s41586-021-04190-y.
[61] M. Li, M. Hilpert, J. Goldsmith, J.L. Brooks, J.A. Shearston, S.N. Chillrud, T. Ali, J. G. Umans, L.G. Best, J. Yracheta, A. van Donkellar, R.V. Martin, A. Navas-Acien, M.-A. Kioumourtzoglou, Air pollution in American Indian versus non-American Indian communities, 2000–2018, Am. J. Public Health 112 (2022) 615–623, https://doi.org/10.2105/AJPH.2021.306650.